Think, Consult, Learn, Reorganize: Recursive Metareasoning for Self-Modifying Agent Societies

Blue Sky Ideas Track

Anonymous Author(s)

Abstract

When an agent is uncertain, should it think longer, consult another agent, acquire a new skill, reorganize its team, or admit that further computation is unlikely to help? Classical metareasoning answers such questions by weighing the expected value of a computation against its cost, but it assumes that the set of available computational actions is known and fixed. Generative agent societies violate this assumption: they can synthesize skills, open communication channels, instantiate specialists and entire sub-societies, and thereby change the very action space over which future metareasoning operates. We propose *Uncertainty-Routed Recursive Agent Societies* (URRAS) as a research agenda for metareasoning in such self-modifying systems. URRAS organizes computation around four escalating interventions—*Think, Consult, Learn, Reorganize*—and lets hard subproblems recursively instantiate child societies governed by the same controller. Uncertainty is treated not as an objective to minimize but as diagnostic evidence for estimating the value of further computation. This perspective exposes new multi-agent problems: identifying *why* a society is uncertain, allocating budgets across nested societies, assigning credit across recursive interventions, eliciting truthful uncertainty reports from strategic agents, and keeping self-modifying organizations stable. We outline these challenges and state falsifiable predictions by which the paradigm can be tested under matched inference budgets.

Keywords

Metareasoning; Value of Computation; Multi-Agent Organization; Recursive Agent Societies; Uncertainty; LLM Agents; Mechanism Design

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1 Introduction: What Should Think Again?

Language-model agents are now scaled along several axes at once: deeper internal reasoning [2, 21], more agents and debate rounds [3, 29], growing skill libraries [23–25], and searched or learned interaction topologies [26, 27]. Each axis has its own controller, typically tuned in isolation. Yet when a society of agents is stuck on a problem, the decisive question is not *how much* more computation to spend but *what kind*. A mathematical derivation that is almost right needs another reasoning pass; a multi-hop question whose evidence sits with a colleague needs a message; a recurring failure on one task family needs a new skill; and a skill borrowed by every agent, every day, may need to become a permanent specialist role.

Choosing among computations by their expected value is an old idea. Rational metareasoning [7, 17], bounded optimality [18], anytime algorithms [28] and, in multi-agent systems, meta-level control [16] all ask which deliberative or coordinative action is worth its cost. We do not claim to have rediscovered this question. We claim that *generative agent societies break one of its central assumptions*. Classical formulations select from a set of computational actions $\mathcal{A}$ that is given in advance. A generative society, by contrast, can write a new skill, spawn a new agent, rewire who talks to whom, or delegate a subproblem to a freshly created sub-society. Each of these actions *changes the action set available to all future metareasoning*. Moreover, the signals metareasoning relies on—confidence, disagreement, predicted gain—are now produced by generative models that are often miscalibrated [12] and, in open societies, possibly strategic.

Thesis. Future agent societies need a theory of *recursive metareasoning*: rational allocation of computation when (i) the space of computations is generated and modified by the society itself, (ii) the controller may invoke copies of itself on subproblems, and (iii) the evidence used for control comes from fallible, possibly self-interested agents. We call the resulting paradigm URRAS and organize it around an escalation vocabulary that we hope is useful in its own right: **Think → Consult → Learn → Reorganize**.

Contributions. This paper offers (1) a formal framing of agent societies whose metareasoning action space is state-dependent and self-modifying (§2); (2) a genuinely recursive operation in which sub-societies run the same controller (§2); (3) an argument that uncertainty must be evidence for, not the objective of, recursive control, with a two-timescale treatment of structural change (§3); (4) seven open problems, several of which are native to the AAMAS community (§4); and (5) concrete, falsifiable predictions that make the vision empirically testable (§5).

2 Recursive Agent Societies

Society state. At task step t a society is

$$\mathcal{S}_t = (N_t,\ G_t,\ K_t,\ M_t,\ \Pi_t), \tag{1}$$

where N_t is the agent population, G_t the interaction graph, K_t the collective skill library, M_t shared memory, and Π_t the metareasoning controller. Given input x_t, the society produces an answer and

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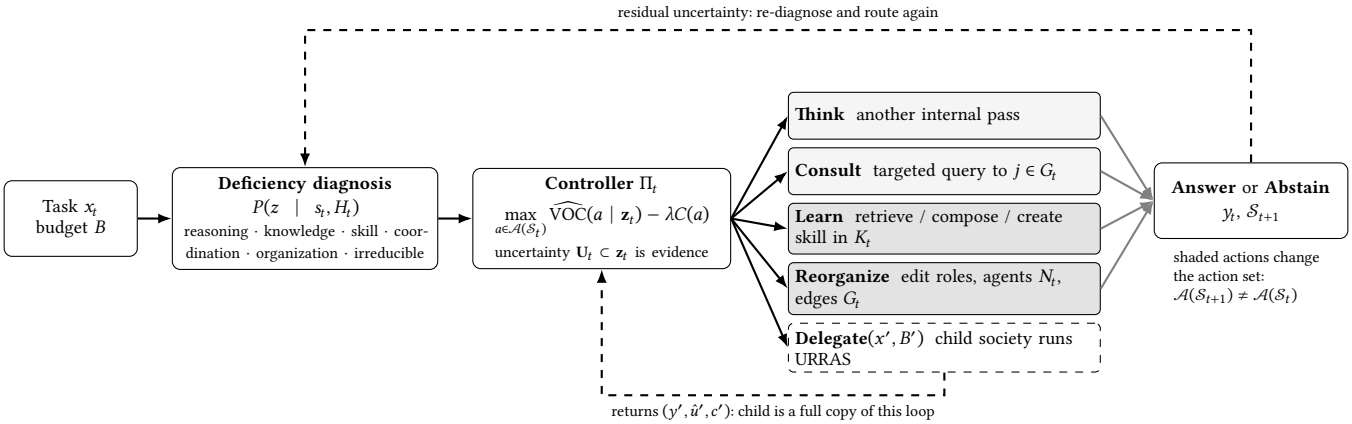


Figure 1: URRAS. The society first diagnoses *why* it is uncertain, then the controller compares the estimated value of each currently available computation against its cost. Learn and Reorganize (shaded) rewrite the action set available to future decisions; Delegate recursively instantiates a child society running the same diagnose–control loop on a subproblem with a budget carved from the parent's.

its own successor state:

$$(\mathcal{S}_t, x_t) \xrightarrow{\Pi_t} (y_t,\ \mathcal{S}_{t+1}). \tag{2}$$

A self-modifying action space. At each internal decision point the controller selects $a \in \mathcal{A}(\mathcal{S}_t)$, where

$$\begin{aligned}\mathcal{A}(\mathcal{S}) = \{&\textsc{Answer}, \textsc{Abstain}\} \cup \mathcal{A}_{\textsc{think}} \cup \mathcal{A}_{\textsc{consult}}(G) \\ &\cup \mathcal{A}_{\textsc{learn}}(K) \cup \mathcal{A}_{\textsc{reorg}}(N, G) \cup \mathcal{A}_{\textsc{delegate}}.\end{aligned} \tag{3}$$

Think actions run further internal reasoning (another pass, a longer chain, a looped-model recursion [2]). Consult actions send targeted queries along edges of G; which partners exist depends on the current graph. Learn actions retrieve, compose, refine or create skills in K. Reorganize actions add or remove edges, reassign roles, or spawn and retire agents. The crucial property is

$$\mathcal{A}(\mathcal{S}_{t+1}) \neq \mathcal{A}(\mathcal{S}_t) \quad \text{in general,} \tag{4}$$

because Learn and Reorganize actions change K, N and G, which index the action families above. A new skill is a new Think or Consult option; a new specialist is a new consultation partner. Classical metareasoning optimizes over a fixed menu; URRAS optimizes over a menu the society is simultaneously rewriting.

Genuine recursion: delegation to sub-societies. We use "recursive" in a strict sense. The action $\textsc{Delegate}(x', B')$ instantiates a child society with its own controller and a budget B' carved from the parent's, runs it on subproblem x', and receives back an answer, an uncertainty report and the cost incurred:

$$\textsc{Delegate}(x', B') \ \Rightarrow\ (y', \hat{u}', c') = \mathrm{URRAS}(x', B'). \tag{5}$$

Because the child may itself delegate, the overall computation is self-similar,

$$\mathrm{URRAS}(x, B) = F\big(x,\ \mathrm{URRAS}(x_1, B_1), \ldots, \mathrm{URRAS}(x_k, B_k)\big), \tag{6}$$

with $\sum_i B_i$ bounded by the parent's remaining budget. This connects LLM agent societies to the long MAS tradition of holonic and nested organizations [4, 9], with one important difference: here the nesting is decided online, per problem, by a value-of-computation controller rather than designed in advance. Figure 1 summarizes the architecture.

3 Metareasoning When the Action Space Changes Itself

Uncertainty is evidence, not the objective. An appealing rule is to choose the action that most reduces uncertainty per unit cost. We argue this is a mistake for agent societies. The cheapest way for a group of language agents to reduce disagreement is often to converge, and convergence on a confident but wrong answer through conformity or sycophancy [19] is a familiar failure mode of debate. An uncertainty-minimizing controller would actively select for such echo chambers. URRAS therefore follows the rational-metareasoning tradition [10, 17]: the controller maximizes estimated value of computation,

$$a_t^* = \arg\max_{a \in \mathcal{A}(\mathcal{S}_t)} \Big[\, \widehat{\mathrm{VOC}}(a \mid \mathbf{z}_t) - \lambda\, C(a) \,\Big], \tag{7}$$

where the features $\mathbf{z}_t = (\mathbf{U}_t, H_t, B_t)$ include a vector of uncertainty signals $\mathbf{U}_t$, the interaction history H_t, and the remaining budget B_t. Uncertainty tells the controller *what may be wrong*; the value estimate tells it *whether fixing it is worth the cost.* $C(a)$ scalarizes heterogeneous costs (tokens, latency, messages, agent lifetime) at an exchange rate set by the deployer.

Diagnosis before allocation. Rather than assume that reasoning, capability, communication and organizational uncertainty can be read off directly, URRAS makes their separation an explicit inference problem. Given the current state s_t, the controller maintains a posterior over the *source* of the deficiency,

$$\begin{aligned}P(z \mid s_t, H_t), \qquad z \in \{&\text{reasoning, knowledge, skill,} \\ &\text{coordination, organization, irreducible}\}.\end{aligned} \tag{8}$$

Each source suggests a different family of actions: reasoning deficits suggest THINK; missing knowledge held elsewhere suggests CONSULT; a capability gap suggests LEARN; recurrent coordination failure suggests REORGANIZE. The last category matters most for termination. Following the distinction between epistemic and aleatoric uncertainty [11], if residual uncertainty is irreducible—the question is ambiguous, or the evidence does not exist—no amount of recursion helps, and ABSTAIN or answering with a calibrated uncertainty report is the rational action. This gives the stopping rule a principled trigger rather than a hand-set threshold.

Two timescales: structural actions are investments. Comparing THINK with REORGANIZE by their immediate effect on the current task is a category error. Another reasoning pass pays now and helps now; creating a specialist pays now and may help on hundreds of future tasks. URRAS therefore evaluates actions with a horizon-dependent value,

$$Q(a_t) = -C(a_t) + \mathbb{E}\Big[\sum_{k\geq 0} \gamma^k \, \Delta V_{t+k} \;\Big|\; a_t, \mathcal{S}_t\Big], \tag{9}$$

where $\gamma \approx 0$ recovers the myopic within-task rule of Eq. 7, while structural actions are judged by amortized benefit over the task distribution. Myopic value estimation is a known limitation of classical metareasoning [7, 17]. In a self-modifying society it becomes central, because the most valuable actions are precisely those whose payoff arrives later and through a changed action set. A natural consequence is a separation of timescales: reasoning recursion operates per token or per pass, communication per exchange, skill change per task family, and organizational change only over many tasks.

Design desiderata. Rather than state results that follow immediately from finite budgets, we list properties a URRAS controller should satisfy and which existing systems rarely check: **Boundedness**—every recursion level and every delegated child respects a budget carved from its parent's; **Marginal value**—recursion continues only while estimated marginal value exceeds cost; **Calibration**—routing signals predict the realized benefit of the computation they trigger, not merely its occurrence; **Reversibility**—skill and structural edits follow a propose–validate–commit cycle and can be rolled back; **Truthfulness**—allocation rules do not reward misreported uncertainty; **Stability**—under a stationary task distribution, organization converges rather than oscillating or growing without limit.

4 What We Do Not Yet Know

P1. Deficiency identifiability. Can a society diagnose *why* it is uncertain before choosing how to spend computation? Distinct deficiencies can produce identical surface symptoms: low confidence on a math problem may reflect a reasoning slip, a missing lemma, or a genuinely ill-posed question. Under what conditions are the sources in Eq. 8 identifiable from observable traces, and how much interventional probing (a cheap THINK to test a hypothesis before an expensive REORGANIZE) is worth? Token probabilities and verbal confidence are weak evidence here [12]; semantic dispersion across samples or agents [13] is a promising but incomplete signal.

P2. Strategic uncertainty reports. In open or heterogeneous societies, uncertainty reports are *messages from agents with incentives*. An agent that receives more compute when uncertain may over-report; one that wishes to avoid verification may under-report. Routing on self-reported uncertainty is therefore a mechanism-design problem. Strictly proper scoring rules [5] offer a starting point—rewarding agent i with $S(\hat{p}_i, y) - \lambda c_i$ makes calibrated reporting optimal in isolation—but it is open how such incentives compose with compute allocation, delegation chains, and collusion between agents that share a backbone model.

P3. Budgets and authority in nested societies. When a parent delegates with budget B', how should B' be chosen before the child's difficulty is known, and can a child request more? How should a child's uncertainty report be discounted by the parent, given that the child may be miscalibrated in correlated ways? Who has the authority to reorganize a sub-society: the parent, the child, or neither? These questions connect resource-rational analysis [14] with organizational theory for MAS [9].

P4. Recursive credit assignment. When success follows THINK → CONSULT → LEARN, which intervention deserves credit, and how should the value estimator $\widehat{\text{VOC}}$ be updated when the action set itself has changed since the decision was made?

P5. Emergent organization. Persistent, task-family-specific capability deficits should drive repeated LEARN actions and eventually REORGANIZE actions that crystallize specialists. Do stable divisions of labor emerge from routing pressure alone, without prescribed roles? Conversely, how do we prevent specialization lock-in when the task distribution shifts?

P6. Stability of self-modifying societies. A candidate Lyapunov-style analysis would define an energy $L_t = \sum_z \alpha_z U_z + \lambda C_t$ and seek conditions under which $\mathbb{E}[L_{t+1} \mid \mathcal{S}_t] < L_t$. We pose this as open: because actions change the action space (Eq. 4), standard convergence arguments for fixed-action controllers do not directly apply.

P7. Recursive scaling laws and governance. How does task performance scale jointly with internal depth, number of agents, communication depth, skill plasticity and delegation depth? Which recursive modifications should require external approval, and how can a self-reorganizing society remain auditable?

5 Testable Predictions

A research agenda should say how it could be wrong. The central empirical claim of URRAS can be tested without building the full vision: *a controller that diagnoses deficiencies should allocate qualitatively different computation to qualitatively different failures, at matched cost.* We state this claim as predictions that any implementation should satisfy.

Settings. The cleanest test contrasts tasks whose dominant deficiency differs by construction. On competition mathematics such as MATH [8, 15], failures are predominantly reasoning deficits. On multi-hop question answering such as MuSiQue [22], with supporting evidence distributed across agents, failures are predominantly missing-evidence deficits. All compared systems should share one

backbone model, so that capability differences cannot masquerade as routing differences, and should include *adaptive* baselines—adaptive self-consistency [1] and debate with early stopping on consensus [3]—since outperforming fixed-depth or fixed-round policies alone would prove little.

Predictions. At matched average inference budget, a controller restricted to Think, Consult and lightweight Learn should satisfy four predictions. It should (H1) match or exceed the accuracy of adaptive baselines; (H2) choose Think more often on mathematics and Consult more often on multi-hop questions; (H3) show the same shift *within* each task family—more Think on harder mathematics problems, more Consult on questions with more hops—so that routing cannot be explained by recognizing the benchmark; and (H4) escalate from Think to Consult after unproductive internal recursion rather than repeating Think. The transition matrix $P(a_{t+1} \mid a_t)$ and the calibration of $\widehat{\mathrm{VOC}}$ against realized gains make failure as informative as success: if routing is indistinguishable across deficiency types, diagnosis is not working, and P1 is harder than it looks.

6 Positioning

URRAS sits above several active lines of work, each of which recurses along one axis. Mixture-of-Recursions [2] adapts recursive depth inside a Transformer. RecursiveMAS [29] casts a multi-agent system as a single latent-space recursive computation over fixed collaboration patterns, with a fixed number of rounds. SkillRL [25] co-evolves a skill library with the agent policy, and MetaSkill-Evolve [24] recursively improves the improvement procedure itself through a multi-agent pipeline on a slower timescale. Topology search methods [26, 27] adapt communication graphs, and learned communication gating [20] decides when agents should talk. Test-time compute scaling [21] and adaptive computation [6] allocate internal computation per input. Meta-level control for MAS [16] is the closest classical ancestor, choosing among deliberation and coordination activities under resource bounds. What none of these addresses is the conjunction URRAS targets: choosing *across* these axes, in an action space the society rewrites, with a controller that recurses into sub-societies, using evidence produced by fallible and possibly strategic agents.

7 Conclusion

Metareasoning was formulated for agents that reason about a fixed repertoire of computations. Agent societies built from generative models can rewrite that repertoire, recruit new members, and spawn copies of their own decision process. We have argued that this turns a classical question into a new one—*what does rational metareasoning become when agents can recursively redesign the computation being reasoned about?*—and that answering it requires the multi-agent community's expertise in organization, coordination, incentives and bounded rationality. When an agent society remains uncertain, the question is no longer only whether to think again, but *what* should think again: one agent, the group, its skills, its structure, or a new society created for the purpose.

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